

Nathan Spelts

(425) 408 - 2655 • nathan@spelts.net • Vancouver, WA
nathanspelts.com • github.com/micro-wizard • linkedin.com/in/nathanspelts

SUMMARY

Embedded software engineer specializing in secure firmware, low-level systems, and Rust and C development for cyber-physical systems. Passionate about reliable embedded design, secure boot, and building tools that strengthen the entire development lifecycle.

WORK EXPERIENCE

Hyster-Yale Materials Handling, Fairview, OR

Aug 2023– Present

- Software Engineer II

- Developed embedded C firmware for safety-critical forklift microcontrollers, targeting reliability, deterministic execution, and compliance with industry safety standards.
- Implemented secure boot improvements and contributed to firmware authentication workflows, strengthening protection against tampering and unsafe OTA updates.
- Designed and built a Rust-based CAN logging and analysis tool using C FFI to interface with Vector XL hardware, enabling real-time and offline diagnostic capability.
- Improved internal debugging workflows by creating Python and Rust utilities for log parsing, firmware build artifact automation, and runtime analysis.
- Collaborated with cross-functional electrical, controls, and test engineering teams to deliver robust cyber-physical systems in industrial environments.

Seasalt.AI, Remote

Jul 2022– May 2023

- Data Science Intern

- Improved Spanish and Bahasa Indonesian Kaldi ASR models by over 40% WER reduction through acoustic/LM tuning, pipeline optimization, and dataset refinement.
- Automated multilingual data collection and processing using Python and Bash, scraping and cleaning thousands of hours of captioned audio for model training.

EDUCATION

Georgia Institute of Technology, Remote

Aug 2025– Present

- Master of Science in Cybersecurity – *Cyber-Physical Systems*

- Current GPA: 4.00
- Coursework focuses on embedded and systems security including secure system design, binary exploitation, network security, database security, and applied cryptography.
- Exploring resilience and hardening techniques for embedded Linux, IoT firmware, and cyber-physical systems.

Washington State University, Vancouver, WA

Graduated May 2023

- Bachelor of Science in Computer Science

- GPA: 3.9 / 4.00
- Relevant Coursework: Operating Systems, Cryptography, Embedded Systems, Data Structures, Algorithms, Software Engineering, Assembly, Statistics, and Linear Algebra.

SKILLS

Programming Languages: Rust, C, Python, C++, Lua, Bash, Powershell, Kotlin, Swift, Javascript, Java

Embedded Systems: Secure Boot, Bootloaders, Bare-Metal ARM, RTOS Concepts, GPIO / I2C / SPI / UART / CAN, Memory-Mapped I/O, Firmware Debugging, GDB, QEMU

Tools Technologies: Git, Docker, Kaldi, FFI (Rust/C), CI/CD Automation, Linux, LaTeX